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Paleomagnetic and geochronological studies on Paleoproterozoic diabase dykes of Karelia, East Finland—Key for testing the Superia supercraton

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ABSTRACT

Paleomagnetic results are presented for two Paleoproterozoic mafic dykes in the Taivalkoski area in northern Karelia Province of the Fennoscandian shield where, based on K-Ar data, the crust has seen minimal effects of the otherwise pervasive 1.8–1.9 Ga Svecofennian orogeny. Within this study a new U–Pb baddeleyite age of 2339 ± 18 Ma has been determined for one of the E–W trending dykes (dyke AD13).

The paleomagnetic results show that a strong Svecofennian overprinting is pervasive in the area. Upon thermal or AF demagnetization four remanence directions were obtained. Most typical are the secondary Svecofennian remanence direction A (intermediate down to the NNW) and remanence direction B (intermediate down to the NNE). Component *D* ($D = 115.4^\circ$, $I = 50.5^\circ$, $\alpha_{95} = 2.6^\circ$) yielding a virtual geomagnetic pole (VGP) *D* (Plat = −19.5°N, Plon = 263.3°, $A_{95} = 3.1^\circ$) is obtained from baked rocks for dyke WD, and based on a positive baked contact test is interpreted to represent the primary magnetization dating from about 2.4 Ga. Dyke AD13 carries only secondary *A* and *B* components, its unbaked host migmatites carry reversed *A* (A_R) component, and the baked host rock carries a component *D*′ ($D = 134.5^\circ$, $I = -7.3^\circ$, $\alpha_{95} = 8.8^\circ$), which yields a VGP pole *D*′ (Plat = −20.4°N, Plon = 257.3°, $A_{95} = 7.6^\circ$), possibly representing magnetization at 2.3 Ga.

The new paleomagnetic data from the Karelia Province compared to similar-aged paleomagnetic data from the Superior Province does not support the recently proposed Superia configuration, based upon dyke swarm trajectories.

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1. Introduction

The Paleoproterozoic era from the amalgamation and dispersal of a possible Neoarchean supercratons named Superia, Vaalbara, and Sclavia or supercontinent Kenorland (e.g. Bleeker, 2003) to the formation of the 1.9–1.8 Ga supercontinent Nuna (a.k.a. Columbia, Hudsonland) might represent the first supercontinent cycle. Supercontinent cycles have shown to have intriguing temporal relationships with core, mantle, crust, oceans, atmosphere and early evolution of life (Reddy and Evans, 2009). However, this first supercontinent cycle and moreover the proposed first Neoarchean supercratons are currently lacking in paleogeographic detail. Several recent and ongoing studies have produced high-quality Precambrian paleomagnetic data, and a paleogeography is becoming defined for the interval of 2.7–1.8 Ga. For example the paleomagnetic apparent polar wander (APW) paths for Vaalbara supercraton, being reconstruction of Kaapvaal and Pilbara cratons (de Kock et al., 2009), and Slave craton, joining possibly Dharwar, Wyoming and Zimbabwe cratons as part of Sclavia supercraton until 2.0 Ga (Bleeker, 2003), are beginning to take form (Buchan et al., 2009; Mitchell et al., 2010). Recently published and preliminary new paleomagnetic data show that Superia comprises the Superior craton and it might have included Kola, Karelia, Wyoming, and Hearne, as indicated by the large igneous province (LIP) magmatic "barcode" record (Ernst and Bleeker, 2010). From these the Superior craton has both the most extensive magmatic barcode record and paleomagnetic data set. Based on these Bleeker and Ernst (2006) have presented a model of a long-lived (2.7–2.0 Ga) supercraton Superia where the Superior, Wyoming, Hearne, Karelia and Kola cratons are tightly joined based on matching of two or more coeval 2.5–2.1 Ga dyke swarms on each craton (see also Ernst and Bleeker, 2010). However the data from different cratons

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are sparse and due to the complexity of the overprint patterns on the ca. 2.45 Ga units of the Karelia craton, Bleeker and Ernst (2006) did not include the paleomagnetic information from the Karelia craton in their model. The Karelia craton around the Taivalkoski area both in Finland and Russia has suffered multiple remagnetization events, the most severe one caused by the Svecofennian orogeny at ca. 1.9–1.8 Ga (e.g. Khramov et al., 1997; Krasnova and Gooskova, 1995; Mertanen, 1995; Mertanen et al., 1989, 1999). This remagnetization has been recognized in most Karelia formations and was originally named as component *A* in the 2.45 Ga layered intrusions in Finland (Mertanen et al., 1989). The layered intrusions carried two other overprints, the other one named as component *B*, which was thought to be related to the vaning stage of the Svecofennian orogeny at ca. 1.75 Ga, and the other one component *E* that was regarded to be ca. 2.1 Ga based on apparent polar wander (APW) path. The characteristic remanence component of the layered intrusions, thought to be the primary 2.45 Ga magnetization, was named as component *D* (Mertanen et al., 1989). Later, in 2.44 Ga mafic dykes in Russian Karelia another *D*-like component, but with lower inclination, was revealed and it was named as component *D′* (Mertanen et al., 1999). Since then, a debate has been going on, whether component *D* with higher inclination or component *D′* with lower inclination represents the primary 2.45 Ga remanence in the Karelia craton. This question has importance because component *D′* would place the Karelia and Superior cratons together unlike component *D*. In this paper we use the names *A*, *B*, *D* and *D′* in the same sense as in the papers by Mertanen et al. (1989, 1999).

So far most of the paleomagnetic data (e.g. Mertanen et al., 1999, 2006a,b; Mertanen and Korhonen, 2011) obtained for Karelia negate the tight Superia fit of Karelia and Superior at 2.50–2.45 Ga. In order to get more evidence on the paleoposition of the Karelia craton at 2.5–2.0 Ga, and to test the proposed Superia model (Bleeker and Ernst, 2006; Ernst and Bleeker, 2010) paleomagnetic and rock magnetic studies on several Paleoproterozoic mafic dykes, especially in the Taivalkoski area in northern Finnish Karelia, have been carried out. Herein we also report a U–Pb baddeleyite age for a dyke which also forms one of the key paleomagnetic sites for this study.

2. Geological setting, sampling and measurements

2.1. Geology and sampling

The eastern Fennoscandian shield comprises the Archean basement complex (3.5–2.6 Ga) and the Paleoproterozoic cover (Fig. 1). The Archean continental core of the Fennoscandian Shield was formed at ca. 3.5–3.2 Ga and it can be divided into the Karelia and Kola Provinces. The Archean Belomorian Province is located between these cratonic domains (Gaàl and Gorbatschev, 1987; Bogdanova, 1996; Slabunov et al., 2006; Hölttä et al., 2012). The study area is located in Taivalkoski area in the northern part of the Lentua Complex of the Western Karelia subprovince (Hölttä et al., 2012), consisting of Archean granitoids and greenstone belts, partly covered by Paleoproterozoic sedimentary formations (Fig. 1). This region was selected because regional K-Ar studies on biotite and hornblende, showed that it was one of the few places where Archean ages survived in hornblende, being close to the zircon ages (Kontinen et al., 1992). This observation implies that the degree of Svecofennian overprinting may be less in this region, thus offering the possibility that paleomagnetism could see through the metamorphism. From the Late Archean onwards (since ca. 2.5 Ga) the Belomorian belt in the northeast and the Proterozoic Svecofennian orogen in the southwest were molded against the Karelia craton (Gaàl and Gorbatschev, 1987; Gorbatschev and Bogdanova, 1993; Bogdanova, 1996).

The entire Karelia craton is cut by voluminous NW-, E- and NE-trending dyke swarms and intrusions/sills that extend from Finland to Russia (e.g. Vuollo, 1994; Amelin et al., 1995; Vuollo and Huhma, 2005). An extensive geochronology and geochemistry campaign summarized in Vuollo and Huhma (2005) provided an improved understanding of these Paleoproterozoic mafic dykes (Fig. 1). Dykes can be divided into at least five main groups with approximate ages of 2.45 Ga, 2.32 Ga, 2.2 Ga, 2.1 Ga, and 1.98 Ga (e.g. Vuollo and Huhma, 2005). Subsequent U–Pb dating identified additionally 2.5 Ga dykes in the Vodlozero terrane of the Karelia province in northwestern Russia (Bleeker et al., 2008). A significant sign of the break-up event on a possible Neoarchaen supercraton is the existence of generally NNW-trending 1.98 Ga tholeiitic and Fe-tholeiitic dykes intersecting Archean northern Karelia and Paleoproterozoic Central Lapland. Later, juvenile continental crust was formed in the present southwestern Fennoscandia during the Svecofennian orogeny at 1.92–1.77 Ga (Gaàl and Gorbatschev, 1987; Gorbatschev and Bogdanova, 1993; Lahtinen et al., 2005; Korja et al., 2006). The Archean core and the Paleoproterozoic units of the Karelia Province were for the most part deformed and metamorphosed during this orogeny.

Tens of Paleoproterozoic diabase dykes were sampled from the Taivalkoski area in the Karelia Province in Finland (Fig. 1) but only two of them (WD and AD13) provided possible primary magnetization directions. Host rocks to the dykes are mainly Archean migmatitic tonalite-trondhjemite-granodiorite (TTG) gneisses and these were sampled in several sites for baked contact tests (Everitt and Clegg, 1962). Samples for paleomagnetic study were taken as block and core samples and oriented using magnetic and/or sun compasses. A block sample for geochronology was taken from dyke AD13.

2.2. Paleomagnetic measurements

Paleomagnetic measurements were carried out at the University of Toronto, Canada (UT); at the Solid Earth Geophysics Laboratory of the University of Helsinki, Finland (UH); and at the Paleomagnetic laboratory of the Geological Survey of Finland, Espoo (GTK). Stepwise alternating field (AF) demagnetizations were done using single-axis demagnetizer with maximum field up to 160 mT, coupled with 2G–DC (UH) or 2G-RF (GTK) SQUID magnetometer, AGICO–LDA–3 AF demagnetizer with maximum field up to 100 mT (UH) and Schonstedt SD-1 demagnetizer up to 100 mT (UT and GTK). Stepwise thermal demagnetization was performed using Schonstedt TSD-1 or homemade furnaces (GTK, UH and UT. To isolate different remanence components, standard multicomponent analyzing methods, principal component analysis (Kirschvink, 1980; Leino, 1991), and the intersecting great circle-technique with end-point analysis (Halls, 1976, 1978) were applied to the data. Mean remanence directions and pole positions were calculated using Fisher (1953) statistics. The paleogeographical reconstructions were plotted with the GMAP program (Torsvik and Smethurst, 1999).

The nature of the magnetic carriers was studied by thermomagnetic analysis of selected specimens using the Agico's CS3-KLY-3S Kappabridge system (UH), which measures the bulk susceptibility (k) of the samples during heating up to 700 °C and cooling back to room temperature (in Argon Gas). Curie temperatures were determined using the Cureval 8.0 – program (www.agico.com).

2.3. Geochronology

The sample from the diabase dyke AD13 contains both coarse and fine-grained portions. It was considered whether the coarser material could be xenolithic, and hence being older material picked

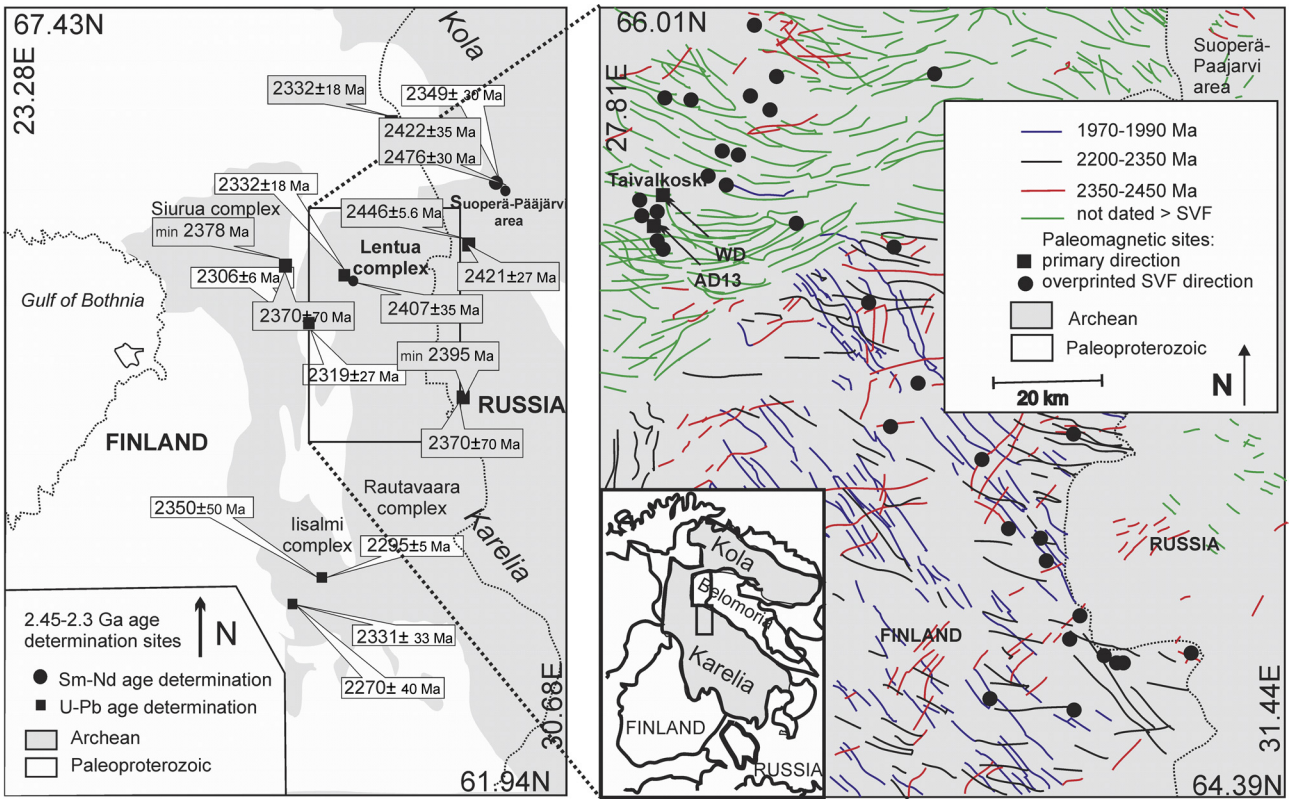


Fig. 1. Previously dated 2.45–2.3 Ga mafic dykes in Finland and a map of the areal distribution of mafic dyke swarms in the Taivalkoski area with the sampling sites for this study (modified from Vuollo and Huhma, 2005; Hölttä et al., 2012). Arrows point to two sites (WD and AD13) that give possibly primary remanence directions (squares). For site AD13 we provide new age data.

up by diabase magma during emplacement. To investigate this possibility we performed XRF analyses (using a hand held instrument, the Thermo Niton XL3t) on both phases, but nothing in that data suggested the coarse and fine material to be of different origins. In any case, the coarse grained material did not yield any baddeleyite, whereas a small fraction of baddeleyite grains (dominated by tiny fragments) were extracted from the finer grained diabase. Baddeleyite was separated using the Söderlund and Johansson (2002) method. The best grains were combined into four fractions comprising 3–4 grains/fragments in each. For most of these dyke samples the baddeleyite crystals were hand selected, washed in warm 4N HNO_3 for 1 h, and cleaned by repeat rinses in Millipore H_2O. The selected grains were loaded into TFE Krogh-type dissolution vessels together with a measured amount of $^{205}Pb/^{233-236}U$ tracer solution and a mixture of HF/HNO_3 (10:1), placed in an oven at 220 °C for at least 36 h, evaporated to dryness and converted to a chloride form by placing in the oven again overnight with 3.1N HCl. Uranium and lead were purified using anion exchange chromatography for fractions weighing > 1 microgram (French and Heaman, 2010) The isotopic composition of U and Pb were determined using a Finnigan TRITON (LIG) thermal ionization mass spectrometer (TIMS) at the Natural History of Museum in Stockholm operating in single collector peak hopping mode. All isotopic data were corrected for mass discrimination, detector bias, spike contribution, blank (1 pg Pb, 0.1 pg U) and initial common Pb (Stacey and Kramers, 1975). The total uncertainty for each analysis was determined by numerically propagating all known sources of error. Age calculations and plots were prepared using the Isoplot software of Ludwig (2003) with the ^{238}U and ^{235}U decay constants and $^{238}U/^{235}U$ value reported by Jaffey et al. (1971).

2.4. Previous isotopic ages

An extensive geochronology and geochemistry campaign on Paleoproterozoic dykes in Karelia is summarized by Vuollo and Huhma (2005). Fig. 1 presents ca. 2.45 Ga and 2.33 Ga isotopic ages from the Karelia Province (Vuollo and Huhma, 2005). The following ages were previously determined from the Taivalkoski area in the Lentua complex: U–Pb: 2306 ± 6 Ma, 2319 ± 27 Ma, 2332 ± 18 Ma, and Sm–Nd: 2407 ± 35 (obtained from dyke WD which provides a possible primary remanent magnetization direction of this study); from the Iisalmi complex: U–Pb: 2295 ± 5 Ma, 2331 ± 33 Ma, and Sm–Nd: 2270 ± 40 Ma, and 2331 ± 33; from the Suoperä-Pääjärvi area (Russian Karelia): U–Pb: 2421 ± 27 Ma, and Sm–Nd: 2349 ± 30 Ma, 2422 ± 35 Ma, 2446 ± 5.6 Ma, and 2476 ± 30 Ma; and from the Siurua complex: U–Pb: 2370 ± 70 Ma, min 2378 Ma, and 2461 ± 150 Ma.

3. Results

3.1. Rock magnetic results

Based on petrophysical properties the dykes form a fairly coherent group. The Koenigsberger's ratio (Q value – the ratio between the natural remanent magnetization, NRM, and induced magnetization) is ca. 2 indicating that samples represent suitable material for paleomagnetic study. Some of the basement rock samples show Q values below 1 indicating that induced magnetization dominates and that therefore these samples may not record the original magnetization. NRM and susceptibility of baked samples are enhanced compared to unbaked samples (Fig. 2a). Thermomagnetic analyses indicate that heating of the dykes changed the magnetic minerals

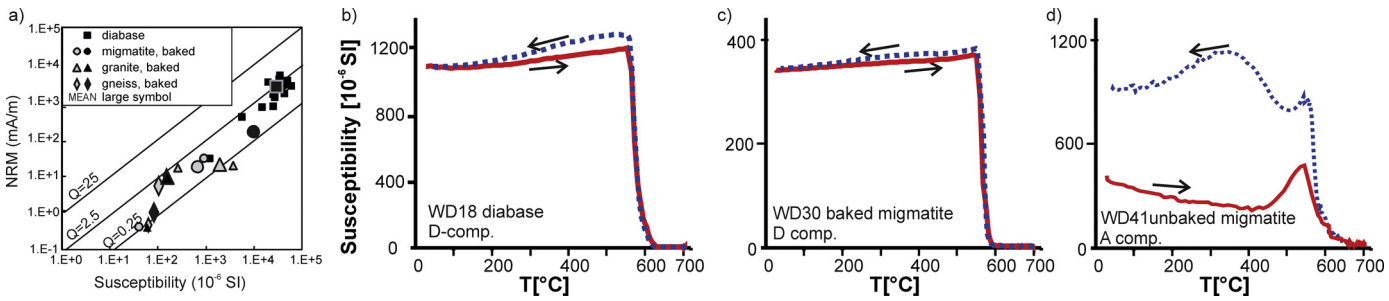


Fig. 2. (a) Petrophysical properties for the studied dykes, baked and unbaked rocks. (b and c) Thermomagnetic curves (susceptibility vs. temperature) for samples from site WD. D (A) comp. – component *D* (A) was obtained during demagnetization. These curves show that due to the heating of the dyke the magnetic minerals in baked migmatite have altered compared to unbaked migmatite.

to more stable magnetite in baked zone (Fig. 2c and d). This is seen in thermomagnetic curves where baked migmatite sample shows reversible heating and cooling curves (Fig. 2c). All studied dyke, and baked host rock samples show Curie temperatures appropriate for (titano)magnetite (550–580 °C). Dyke and baked host rock sample show a slight Hopkinson's peak before the Curie temperature indicating single-domain (SD) magnetite. The unbaked sample (Fig. 2d) shows a pronounced Hopkinson's peak, but the irreversible curves show that mineralogical changes took place during heating.

3.2. Paleomagnetic results

Multicomponent analyses of paleomagnetic data show that the samples taken from all of the studied dykes carry four remanence components other than viscous magnetization. We identify these as *A*, *B*, *D*, and *D′* (Table 1). Normally, each specimen carries no more than two components. Results obtained from Taivalkoski add to those obtained from NW–SE or E–W trending, subvertical Paleoproterozoic dykes in Russian Karelia at Lake Pääjärvi and Suoperä areas (Mertanen et al., 1999) and the Vodlozero terrane (Burakovka intrusion and Shalskiy dyke; Khramov et al., 1997; Mertanen et al., 2006a). The majority of all dykes show a NW intermediate-down directed component A that was also clearly separated from the dyke AD13 and as a low coercivity component from few specimens from dyke WD (Figs. 3–6). Component A has been interpreted to be caused by the Svecofennian orogeny (e.g. Mertanen et al., 1999 and references there in). Almost as common in all studied dykes is a NNE intermediate-down directed component B that was separated at low coercivities and low unlocking temperatures from dyke WD (Fig. 4 and Fig. 6a). In majority of the dykes the direction of B component is very close to direction of component A and both components have similar, overlapping coercivity and unblocking temperature spectra. However, some of the dykes show both A and B as distinct components. Usually coarser-grained samples from dyke interiors show component B and finer grained samples, closer to the contact, show component A, suggesting that component B is younger than A (Table 1).

3.2.1. Dyke WD, component D

Although the majority of dykes contain only components A and B, two dykes WD and AD13 which are well-exposed in road-cuts along the Taivalkoski Road, reveal also other components. In dyke WD an intermediate down to the E-ESE component *D* is observed and is also found in a thin dyke that crosscuts dyke WD and in baked host rocks. Dyke WD has a Sm–Nd model age of 2407 ± 35 Ma with ε_{Nd} + 1.6 (Vuollo and Huhma, 2005). The Sm–Nd age can be questionable, but in several cases in the Karelia craton where the dyke rock has both U–Pb and Sm–Nd ages, these ages agree (Vuollo and Huhma, 2005). According to thin section studies, also in dyke WD the fresh looking pyroxene and plagioclase grains have crystallized simultaneously in a closed system which gives support that the Sm–Nd model age is close to the original cooling age.

Component *D* is separated in a narrow temperature range of 520–560 °C, but has a coercivity spectrum (30–100/160 mT) as wide as that for component *B*. In dyke WD component *B* moves toward component *D* during both thermal and af demagnetization (Fig. 4), but seldom is a stable end point achieved. Such behavior makes it hard to separate *B* and *D* components from WD dyke samples. It is possible that much of the movement is actually between the true *D* component and component *A* or the Present Earth's field (PEF) direction. However component *B* is obtained in many areas across the Fennoscandian shield, where component *D* is not, which indicates that it has a geological origin. Baked basement rock samples (migmatites) for dyke WD show only component *D* and a viscous PEF component (Fig. 3). Because of the clearer distinction between

Table 1
Mean paleomagnetic data for Taivalkoski dykes and host rocks.

	Component	Str/dip (°)/(°)	width (m)	*N*/*n*	*D* (°)	*I* (°)	α_{95} (°)	*k*	VPGlat (°)	VPGlon (°)	A_{95} (°)	*K*
Site WD Lat 65.58°N, Lon: 28.87°E												
WD diabase dyke	*D component*	95/90	12	28*/56	76.9	62.2	3.1	39.1	45.5	107.5	3.9	24.2
Dyke cross cutting WD	*D component*	90/90	0.1	4*/7	84.8	59.8	12.5	24.2	41.4	103.3	16.5	14.4
WD bakedmigmatite	*D component*			27*/94	115.4	50.5	2.6	31.8	−19.5	263.3	3.1	23.5
WD diabase dyke (low coercivity overprint)	*B component*	95/90	12	25*/45	15.0	40.4	14.5	5.0	48.9	187.3	13.5	5.6
Site AD Lat 65.52°N, Lon: 28.05°E												
AD13 dyke	A_N *component*	95/90	38	18*/35	343.7	41.0	3.2	59.7	47.0	230.3	3.2	58.2
AD13 baked migmatite	*D′ component*			1/9*	134.5	−7.3	8.8	35.2	20.4	257.3	7.6	46.9
AD13 Unbakedmigmatite	A_R *component*			3*/4	150.1	−52.7	14.3	74.9	−53.8	72.1	14.9	69.6

Lat/Lon the latitude and longitude of the sampling site. *N*/*n* number of samples/specimens. *D* and *I* the declination and inclination of the remanent magnetization.k the best estimate for precision parameter of Fisher (1953). α_{95} radius of circle of 95% confidence of direction. VGPlat, VGPlong the latitude, longitude for the virtual geomagnetic pole. A_{95} radius of the circle of 95% confidence of the pole. *K* is the best-estimate of the precision parameter κ for the observed distribution of site-mean VGPs.

* Statistical level used for mean calculations.

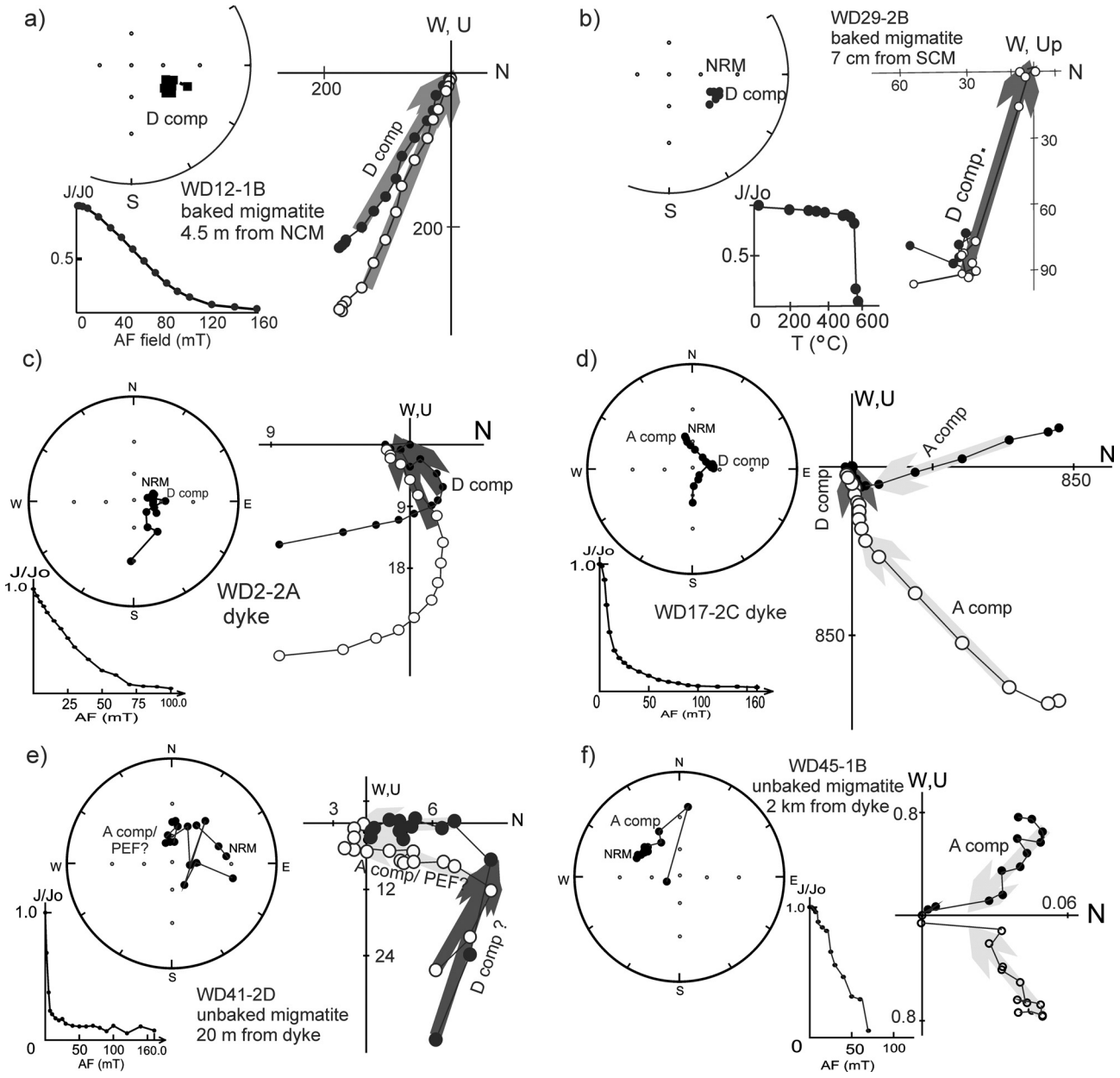


Fig. 3. Examples of demagnetization behavior of dyke and baked host rock samples for site WD. In (a)–(f) upper left is stereoplot, lower left is intensity decay curve, and right is orthogonal (Zijderveld, 1967) demagnetization diagram, where open (closed) symbols denote vertical (horizontal) planes. Numbers are values of magnetization in mA/m.

Table 2
U–Pb TIMS data.

Analyses no. (number of grains)	U/Th	Pbc/Pbtot[a]	$^{206}Pb/^{204}Pb$ Raw[b]	$^{207}Pb/^{235}U$ [corr][c]	±2s %err	$^{206}Pb/^{238}U$ [corr][c]	±2s %err	[Age, Ma] $^{207}Pb/^{235}U$	$^{206}Pb/^{238}U$	$^{207}Pb/^{206}Pb$	±2s %err	Concordance
Bd-a (5 grains)	7.2	0.195	289.3	8.3380	2.06	0.41272	1.96	2268.5	2227.4	2305.7	10.6	0.966
Bd-b (2 grains)	7.3	0.258	231.4	9.0894	3.53	0.44312	3.26	2347.0	2364.6	2331.8	21.6	1.014
Bd-c (2 grains)	3.4	0.143	373.3	8.8944	1.54	0.43305	1.43	2327.2	2319.4	2334.1	9.6	0.994
Bd-d (3 grains)	5.8	0.140	456.1	7.6456	2.45	0.38979	2.29	2190.2	2121.8	2254.9	14.2	0.941

[a] Pbc = common Pb; Pbtot = total Pb (radiogenic + blank + initial).
[b] Measured ratio, corrected for fractionation and spike.
[c] Isotopic ratios corrected for fractionation (0.1% per amu for Pb), spike contribution, blank (1 pgPb and 0.1 pg U), and initial common Pb. Initial common Pb corrected with isotopic compositions from the model of Stacey and Kramers (1975) at the age of the sample.

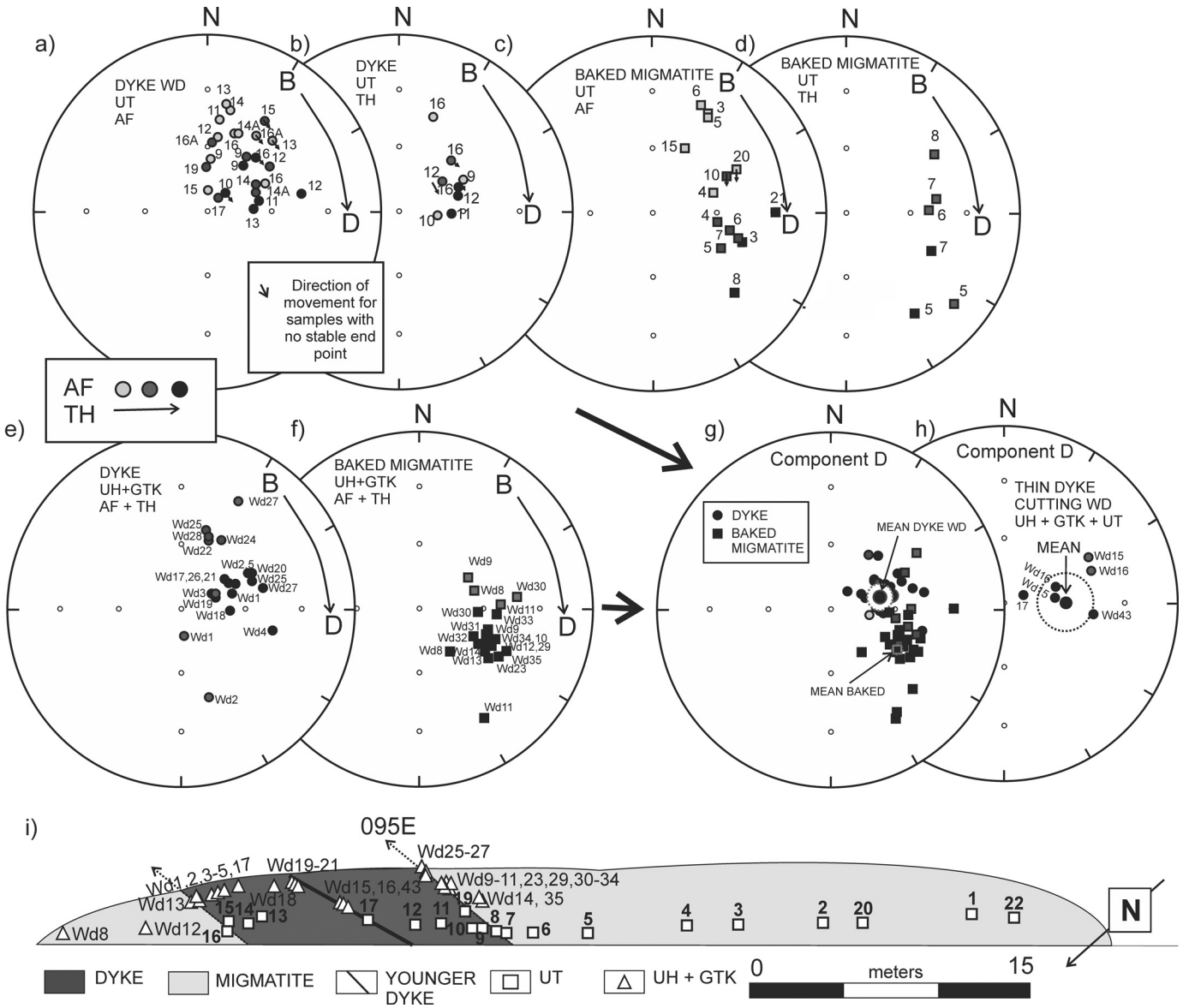


Fig. 4. (a)–(d) Remanence directions for samples from site WD including low coercivity and low temperature components (UT data). (e) and (f) characteristic remanent magnetization (ChRM) directions for samples from site WD (UH + GTK data), (g) mean remanence directions and α_{95} confidence circles of ChRM for site WD, (h) mean remanence directions and α_{95} confidence circles for thin diabase dyke crosscutting the WD dyke (UH + GTK + UT data). Closed (open) symbol denotes downward (upward) direction of magnetization. B (D) indicates the direction of component B (D). (i) Sketch showing sampling at site WD. UH + GTK – samples taken by LJP and SM that have been measured at the University of Helsinki and GTK are marked with two letters (Wd) and numbers, UT – samples taken by HH and JV that have been measured in University of Toronto are marked only with numbers. AF – alternating field demagnetization, TH – thermal demagnetization. Note the color coding for (a)–(f): black indicates the highest, dark gray indicates intermediate, and light gray indicates lowest unblocking temperature and coercivity. If there are several symbols with different shades of gray for one specimen it indicates that several components with different unblocking temperature (coercivities) were obtained for this specimen.

coercivity spectra and unblocking temperatures we interpret that baked migmatites reveal the original direction for component *D* ($D = 115.4^\circ$, $I = 50.5^\circ$ with $\alpha_{95} = 2.6^\circ$). For dyke WD we observe the direction of $D = 76.9^\circ$, $I = 62.2^\circ$ with $\alpha_{95} = 3.1^\circ$ and for crosscutting thin (10 cm) diabase dyke $D = 84.8^\circ$, $I = 59.8^\circ$ with $\alpha_{95} = 12.5^\circ$. The remanent magnetization of the unbaked basement rocks for site WD is weak and unstable, but clearly different from the baked basement rocks.

3.2.2. Dyke AD13, component D′

A shallow SE-directed component *D′* is observed only for the baked basement rocks for dyke AD13 (Fig. 5a and b). We separated a shallow (up, down) SE directed component *D′* at the baking zone (as far as 9 m from the 22 m wide dyke) and at the contacts for the dyke AD13 whereas the dyke itself only shows component *A* (Fig. 5c). Unbaked basement samples for this site were often unstable, but the basement rock beyond the baking zone gives clearly a different stable direction which is antipodal to component *A* (Fig. 5d and Fig. 6). This is one of the few cases where a reversed Svecofennian direction (component A_R) is obtained. This raises concern since in Fig. 5a) and b) the Zijderveld tracks are slightly but clearly curved and show an almost Svecofennian direction in low coercivities. In Fig. 5(c) and (d) with Svecofennian component A_N or A_R the tracks go straight to the origin. This raises a question if it is then possible that component *D′* at site AD13 could actually be a composite of A_N or A_R. We think that the answer is found in rock magnetic measurements, which show that baking by dyke AD13 at the baking zone produced stable magnetite that shows reversed

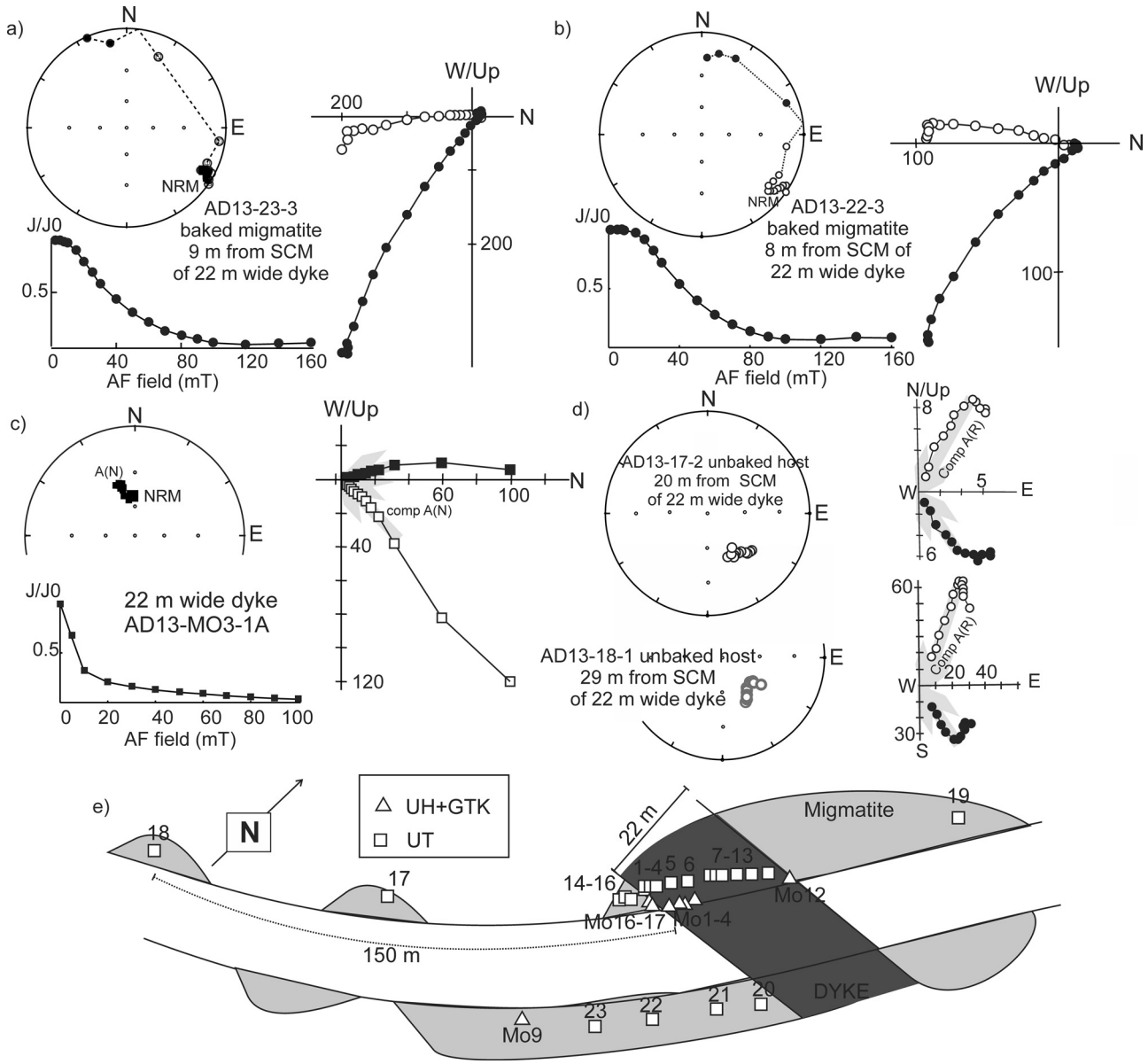


Fig. 5. Examples of demagnetization behavior of dyke and host rock samples for site AD13. In (a)–(d) upper left is stereoplot, lower left is intensity decay curve, and right is orthogonal demagnetization diagram, where open (closed) symbols denote vertical (horizontal) planes and numbers are values of magnetization in mA/m. Note the change in axis between (a) to (c) and (d). (e) Sketch showing site AD13. UH + GTK – samples taken by LJP and SM have been measured at the University of Helsinki and the GTK, UT – samples taken by HH and JV have been measured in the University of Toronto.

heating and cooling curves during the thermomagnetic measurements. These changes render remanence resistant to change during the subsequent Svecofennian orogeny which only affected the dyke interior and basement rocks beyond the baked zone, by remagnetizing them with component A. However, we are aware of that these results do not fulfill the requirements for full positive baked contact test. Anyhow, since this is the first time that component D' ($D = 134.5^\circ$, $I = -7.3^\circ$, $\alpha_{95} = 8.8^\circ$) is observed in the Karelia Province in Finland and may represent a primary component, U–Pb geochronology to obtain its age was carried out for the dyke AD13.

3.3. Geochronology

Results from four separated baddeleyite fractions are shown in Fig. 7 and in Table 2. Due to the small amount of sample the precision in age is relatively poor, but 2 of 4 fractions are concordant and a regression using all 4 fractions yields an upper intercept age 2339 ± 18 Ma. Alternatively, the more precise concordia age of 2333 ± 10 Ma, based just on the concordant fractions alone, can be interpreted as the age of this sample.

4. Discussion

4.1. Poles

The new U–Pb baddeleyite age of 2339 ± 18 Ma has been determined for an E–W trending dyke (site AD13) in the Taivalkoski area of Karelia Province. This adds to the evidence for a significant pulse of 2.33 Ga dyke magmatism in the eastern and northern parts of the Fennoscandian Shield (see Section 2.4). However, based on geochemical data low-Ti tholeiitic ca. 2.45 Ga–1.98 Ga dykes do not form distinct compositional groups based on age (Vuollo and Huhma, 2005).

Table 3
Selected poles for Karelia (Kola) and Superior cratons.

Code	Site	Age (Ma)	Method	Plat (°N)	Plon (°E)	A_{95} (°)	Van der Voo 123456 Σsum	Rplat	Rplon	Poleoragereference
	KARELIA (KOLA)									
Shal	Shalskiythickgabbronoritedyke	2510.6 ± 1.5	U–Pb	22.7	222.1	11.5	101110 Σ4	−46.8	291.5	Mertanen et al. (2006a) and Bleeker et al. (2008)
Paa D′	Karelian baked basement (Lake Pääjärvi, D′)			9.6	256.2	6.7	001010 Σ2	−27.8	329.5	Mertanen et al. (1999)
Bura centr	Burakovka (centralblock)	2449 ± 1	U–Pb	−27.6	260.0	9.2	101010 Σ3	9.0	322.5	Fedotova et al. (1999) and Amelin et al. (1995)
Buradykes	Burakovkadykes (Avdeev-gabbronorite&Shalskiy diabase)			−12.3	243.5	14.0	011110 Σ4	−9.0	310.9	Mertanen et al. (2006a)
Paa D	Russian Karelia maficdykes (Lake Pääjärvi &Suoperä D)	2446 ± 5	U–Pb	−19.9	278.7	6.1	111010 Σ4	7.0	341.3	Mertanen et al. (1999) and Vuollo and Huhma (2005)
Koil	Koillismaa layeredintrusion (D)	2436 ± 5	U–Pb	−29.8	269.3	22.0	101010 Σ3	13.6	329.5	Mertanen et al. (1989) and Alapieti (1982)
TK D	Taivalkoski WD dyke (D) VGP	2407 ± 35	Sm–Nd	−19.5	263.3	3.1	011110 Σ4	3.0	327.3	This work and Vuollo and Huhma (2005)
TK D′	Taivalkoski AD13 dyke (D′) VGP	2333 ± 10	U–Pb	20.4	257.3	7.6	101100 Σ3	−37.7	334.7	This work
Kuet	KuetsyarviFrm. (Belomoria + Kola) SUPERIOR	2058 ± 2	207Pb/206Pb	24.7	300.8	16.7	101110Σ4	24.9	199.3	Torsvik and Meert (1995) and Melezhik et al. (2007)
PtMis	Ptarmigan-Mistassini	2505 ± 2	U–Pb	−45.3	213	13.8	101010Σ3			Fahrig et al. (1986), Buchan et al. (1998) and Evans and Halls (2010)
Mat R	Matachewan R	2473–2446	U–Pb	−44.1	238.3	1.6	111110Σ5			Halls and Davis (2004)and Evans and Halls (2010)
Mat N	Matachewan N	2446 ± 3	U–Pb	−52.3	239.5	2.4	111110Σ5			Halls and Davis (2004)and Evans and Halls (2010)
Nip	Nipissing N1 (B)	2217 ± 4	U–Pb	−17	272	10	111111 Σ6			Buchan et al. (2000)
Sen	Senneterre (B)	2216 + 8/-4	U–Pb	−15.3	284.3	6	111111Σ6			Buchan et al. (1993)
Bis	Biscotasing (N)	2172–2167	U–Pb	26	223.9	7	111110Σ 5			Buchan et al. (1993), Halls and Davis (2004)and Halls et al. (2005)
Mar N	Marathon N	2126–2121	U–Pb	54.1	188.9	7.7	111010 Σ 4			Buchan et al. (1996), Hamilton et al. (2002)and Halls et al. (2008)
Mar R	Marathon R	2106–2101	U–Pb	63.8	168.9	7.5	111110 Σ 5			Buchan et al. (1996), Hamilton et al. (2002)and Halls et al. (2008)
C.Lak	Cauchon Lake (R)	2091 ± 2	U–Pb	62.4	167.3	7.7	111110 Σ 5			Halls and Heaman (2000)and Evans and Halls (2010)
F.Fr	Fort Frances (R)	2076 + 5/−4	U–Pb	51.5	172.7	6.1	111010 Σ 4			Halls (1986), Buchan et al. (1993), Buchan et al. (2007), Wirth et al. (1995), Hamilton et al. (2002)and Evans and Halls (2010)
L.Es	Lac Esprit (R)	2069 ± 1	U–Pb	62	170.5	6.4	111010 Σ 4			Buchan et al. (2007)and Evans and Halls (2010)

Code – code in Figs. 3, 10 and 11; Plat – pole latitude; Plong – pole longitude; A_{95} – Fisher (1953) statistics parameter.Van der Voo (1990) reliability criteria for paleomagnetic data.RPlat and RPlon – Karelia and Kola poles rotated to Bleeker and Ernst (2006) fit by Euler pole of ELat = 75.5°, ELon = 251° and rotation angle = −114°. SVF – Svecofennian.

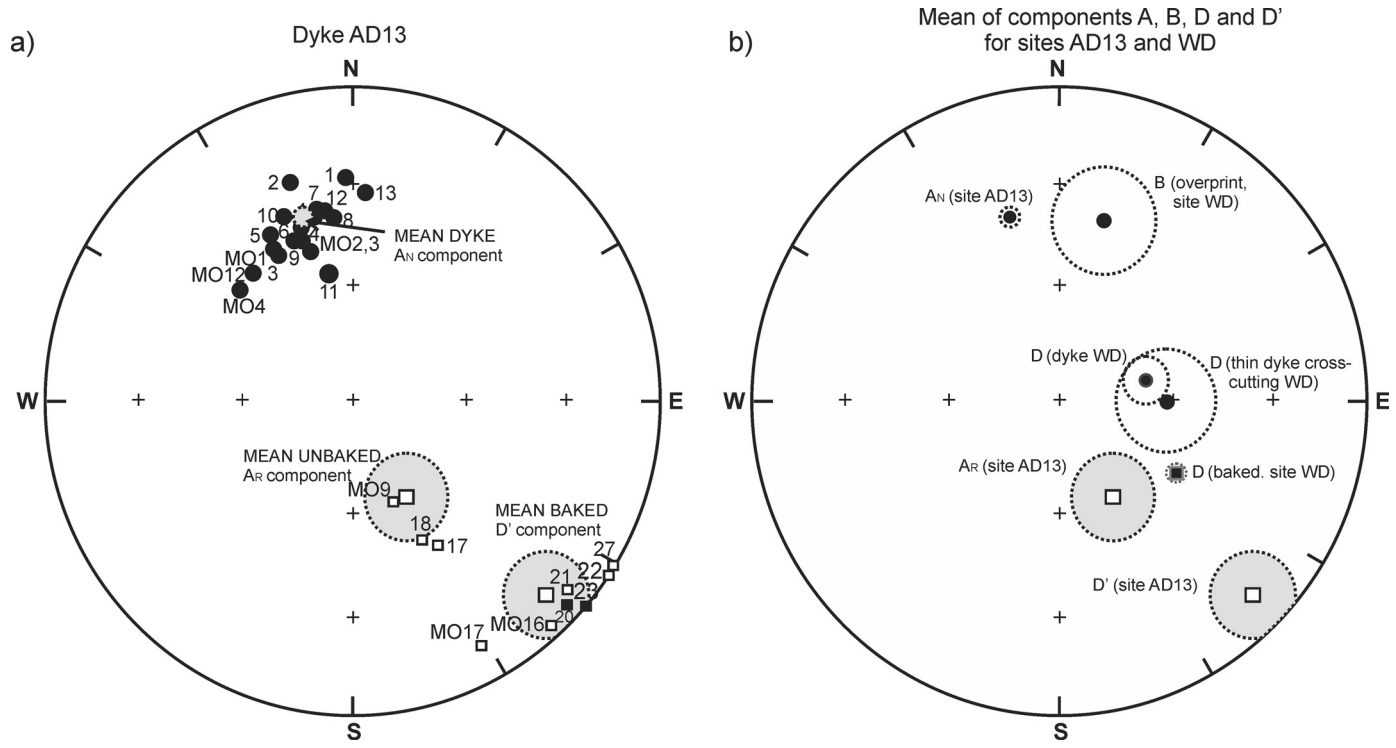


Fig. 6. Mean remanence directions and α_{95} confidence circles for site AD13. Closed (open) symbol denotes downward (upward) direction of magnetization. Some of the samples have been identified with letters AD and numbers (AD-1) these are indicated in figure with numbers (1). Other samples have been identified with letters AD-MO and numbers (AD-MO1) and these are indicated in the figures with MO and numbers (MO1).

Virtual geomagnetic poles (VGPs) calculated for components D′ and D are listed in Tables 1 and 3 and shown in Fig. 8 with other relevant Precambrian poles for Baltica. We obtained a pole *D′* (Plat = 20.4°, Plon = 257.3°, A_{95} = 7.6°) for the baked host rock for the dyke AD13 (2339 ± 18 Ma) that places the Karelia Province on the equator at the time of remanence acquisition (Fig. 9). A corresponding low inclination magnetization direction yielding a pole Plat = 10°N, Plon = 256° (Paa D′ Figs. 8 and 9) was previously obtained in two gabbronorite/Fe-tholeiitic dykes and their basement rocks in the Lake Pääjärvi area, Russian Karelia, and it was suggested that it is one candidate for the primary 2.45 Ga remanence (Mertanen et al., 1999). The confidence circles of VGP D′ of this study (TK D′) and the pole of Lake Pääjärvi (Paa D′) overlap (Fig. 8), so on this basis they may have a similar age. The 2510 Ma (Bleeker et al., 2008) thick Shalskiy gabbronorite close to Burakovka layered intrusion in the Vodlozero terrane shows a similar low inclination component but a declination pointing almost to the south (40° difference to TK D′) leading to different paleomagnetic pole position (Fig. 8). It was interpreted to represent the primary 2.5 Ga remanence and it corresponds to the pole Plat = 23°N, Plon = 222°, A_{95} = 12° (Mertanen et al., 2006a,b; Mertanen and Korhonen, 2011). We have plotted overprint poles A_N and A_R that were obtained from AD13 site on Fig. 8. Pole A_R to poles Paa D′ and TK D′ and further to pole A_N appear to form a swath which leaves the question if the VGP D′ is actually a composite of A_N and A_R.

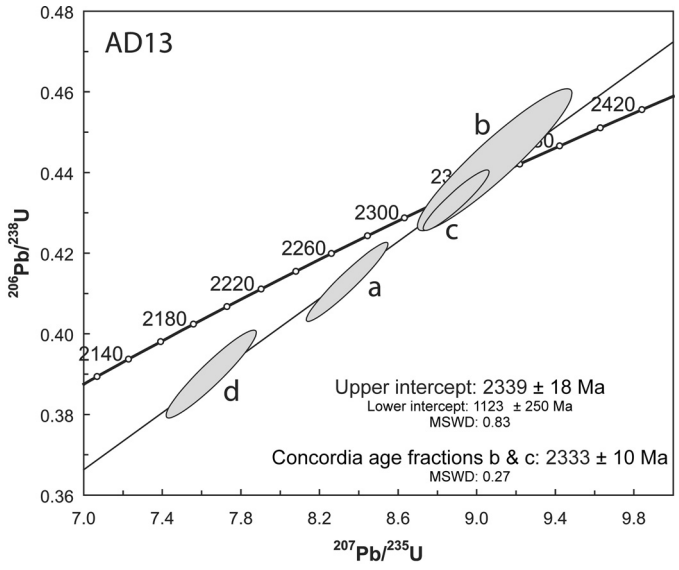


Fig. 7. Concordia diagram.

Baked rocks for dyke WD yield a VGP D (Plat = −19.5°N, Plon = 263.3°, A_{95} = 3.1°) that places Baltica at a latitude of ca. 30° (Figs. 8 and 9). Dyke samples show similar inclinations of ca. 50–60° but ENE declinations instead of ESE pointing declinations (40° difference; Fig. 4), which may be the result of inadequate separation of components *B* (or *A*) and *D*, secular variation during the cooling, or block rotations during the cooling. We favor the first option since the coercivity spectra of components *B* and *D* overlap (see Fig. 4). The overprint components *A* and *B* occur typically in the dyke (Fig. 4), while in many baked host rock samples components *A* or *B* do not occur, but *D* is found as a single component (Fig. 3a and b). This indicates that, like in case of dyke AD13, the dyke was more susceptible for remagnetization than the baked host rock. We also suggest that the overprint component *B* is real and not a combination of the PEF and component *D*, since it is obtained in many locations in the Fennoscandian shield, where component *D* is not found (Mertanen et al., 2008).

The age for the magnetization of component *D* is not clear for two reasons. The first is the lack of robustness of Sm–Nd method used to obtain age (2407 ± 35 Ma) for dyke as discussed above. The second is the fact that a similar component has been previously

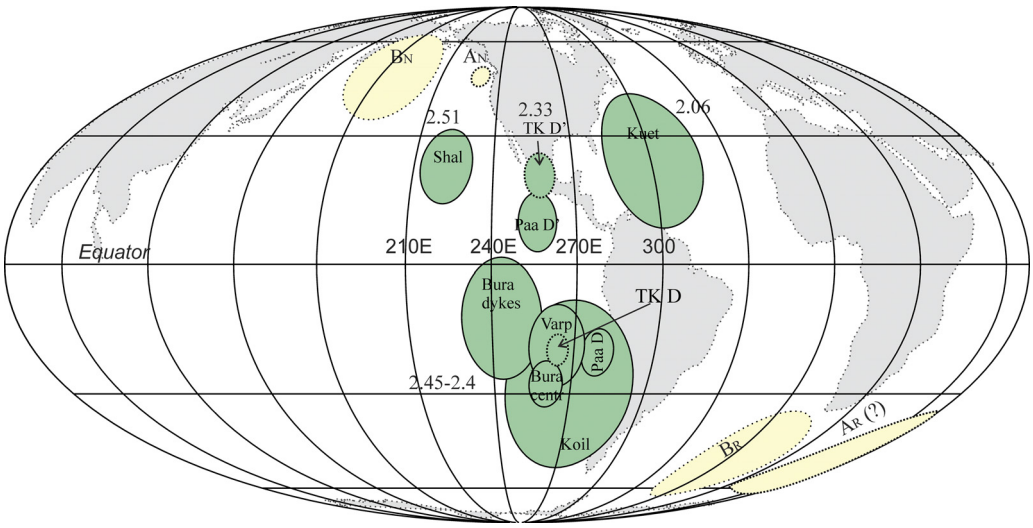


Fig. 8. Taivalkoski virtual geomagnetic poles (VGPs) plotted with relevant 2.5–2.06 Ga poles for Karelia (and Kola, Kuetsyarvi). Used poles are listed in Tables 1 and 3. Pole (VGP) A_N is Svecofenninan overprint on dyke AD13 samples. Pole (VGP) A_R is obtained only from unbaked basement samples for dyke AD13 and is interpreted to represent rarely obtained reversed magnetization due to Svecofennian overprinting event. Pole (VGP) B_N is a low coercivity overprint obtained from dyke WD and B_R is inverted B_N.

obtained also for younger formations, like for 2295 ± 5 Ma (Hölttä et al., 2000) Tulisaari dyke in Varpaisjärvi area in central Finland (Mertanen et al., 2006b), for 2058 ± 6 Ma Kuetsyarvi formation in Kola Province (Torsvik and Meert, 1995) and for 1.97 Ga Konchozero sill in nortwestern Russian Karelia (Pisarevsky and Sokolov, 1999). There is extensive discussion of the possible origin of component D in Mertanen et al. (1999, 2006a,b). In Kuetsyarvi formation the similar component was regarded as an overprint and interpreted to represent a later Ediacaran remagnetization (Torsvik and Meert, 1995). However, if the *D*-like componet would be an Ediacaran remagnetization one would expect to see it also on other Archean-Paleopropterozoic formations all over the craton. Likewise, in the isotopic age determinations there would be some indications of such a young overprint, and this has not been observed. The Taivalkoski *D* pole is in close agreement with the poles obtained for 2.45 Ga gabbronorite dykes in Lake Pääjärvi area (Paa D in Fig. 8) in Russian Karelia (Krasnova and Gooskova, 1995; Mertanen et al., 1999), and for 2436 ± 5 Ma Koillismaa layered intrusions (Koil in Fig. 8) (Mertanen et al., 1989). In the Lake Pääjärvi and at Koillismaa areas a heavy Svecofennian overprint also occurs. Similar D-like poles have been obtained also from the 2.44 Ga Burakovka intrusion (Bura centr. in Fig. 8) (Khramov et al., 1997; Fedotova et al., 1999) and from the Avdeev gabbronorite dyke (Bura dykes in Fig. 8) which is geochemically related to the Burakovka intrusion (Mertanen et al., 2006a,b). Both intrusions are in the Vodlozero terrane which was not affected by the Svecofennian overprinting. There the poles have been interpreted to represent ca. 2.45 Ga primary magnetization. However, there are no positive baked contact tests to prove this. Moreover, both in Burakovka and Lake Pääjärvi areas the component D is also obtained from the Archean basement

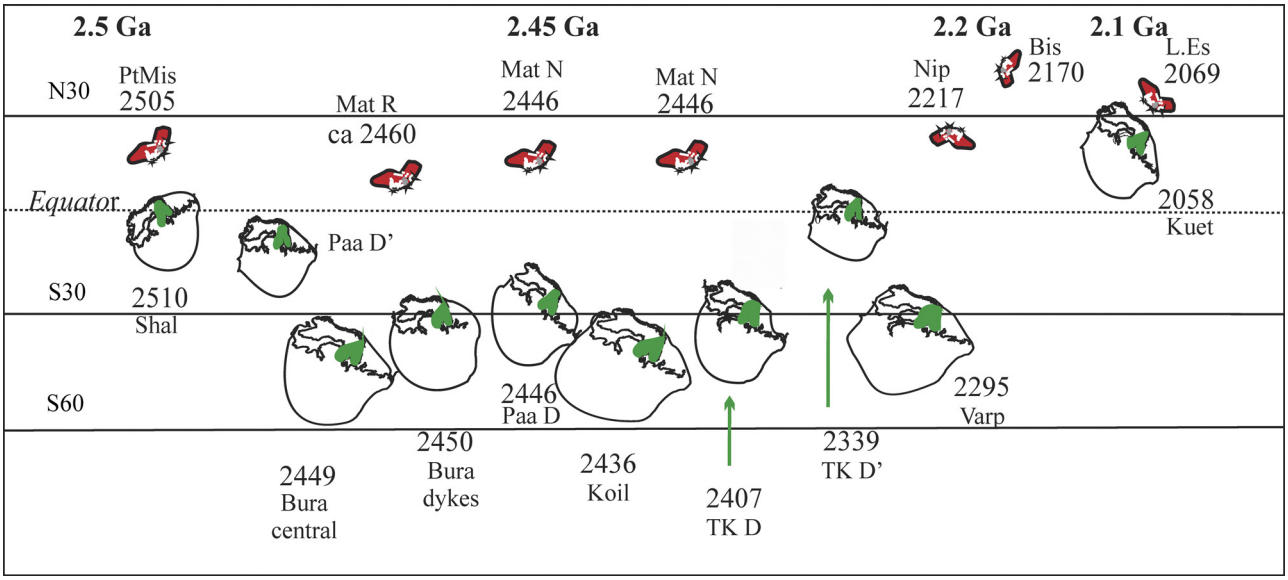


Fig. 9. Cartoon showing latitudinal drifts of Superior (red) and Karelia-Kola (green) within Baltica from 2.5 to 2.1 Ga. Numbers denote U–Pb ages in Ma except in the case for TK D, where the age is modeled Sm–Nd age. Data is listed in Table 3. Note: Time scale shown across top is not in scale. Superior is drawn twice with Matachewan N pole to compare with various 2.45 Ga data from Karelia. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

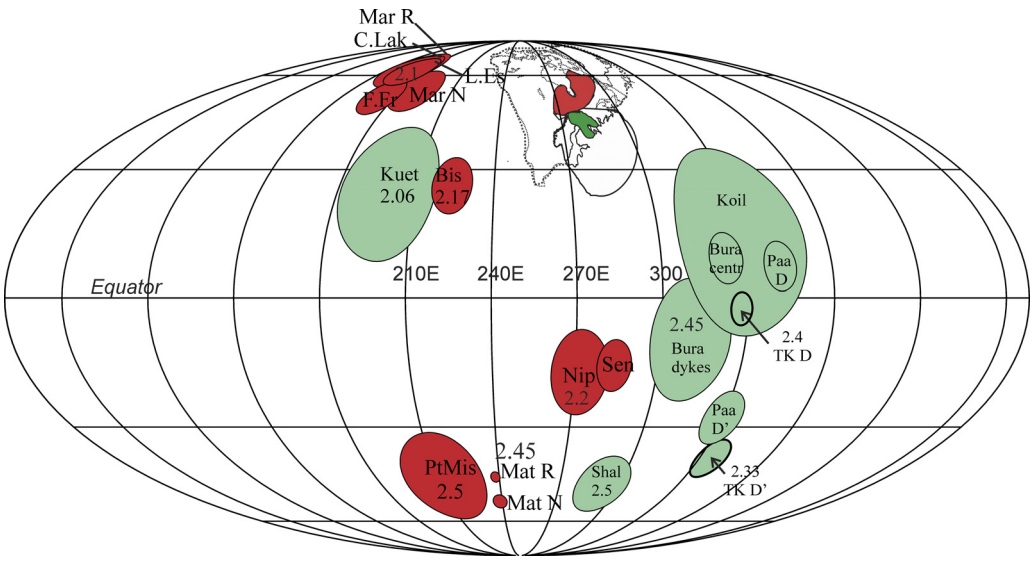


Fig. 10. Superia fit (modified from Bleeker and Ernst, 2006) in present reference frame of Laurentia. Paleomagnetic poles of Karelia and Kola (green in web and light gray in paper print) and Superior (red in web and dark gray in paper print) cratons from 2.5 to 2.0 Ga do not support the proposed fit. Used poles are listed in Table 3. Numbers denote ages in Ga. Karelia (and Kola) poles are rotated to present reference frame of Laurentia by Euler parameters ELat = 75.5°, ELon = 251° and rotation angle = −114°. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of the article.)

(Mertanen et al., 2006a,b) where it was interpreted to be caused by reheating due to extensive 2.45 Ga magmatism in thea areas (Mertanen et al., 2006a,b). In case of the new Taivalkoski data we suggest that magnetization *D* is primary because the baked basement samples for dyke WD show clearly component *D* whereas unbaked do not.

4.2. Proximity of Karelia and cratons in the Paleoproterozoic

Based on a comparison of the trends of 2500 Ma, 2450 Ma, and 2100 Ma swarms in Karelia and Kola Provinces and southern Superior Province, northern Karelia has been placed against the southern Superior Province, in the proposed supercraton Superia (Bleeker, 2003). Based on matching "magmatic barcodes" other cratons in supercraton Superia may have been Hearne, and Wyoming (Ernst and Bleeker, 2010). Mafic dykes on Karelia, Kola and Superior Provinces also share similar geochemistry (Vuollo et al., 1995). Further support for the close proximity of these cratons comes from the occurrence of ca. 2.0 Ga ophiolites in both Fennoscandian and Canadian Shields (Vuollo et al., 1995; Lahtinen et al., 2008; Ernst and Bleeker, 2010).

To study the suggested close proximity of the Karelia and Superior cratons during 2.5–2.1 Ga, we have compared coeval paleomagnetic poles of 2.5 Ga, 2.45 Ga and 2.1 Ga from both Karelia and Superior cratons (Fig. 9). We tested the proposed Superia fit of Bleeker and Ernst (2006) by rotating paleomagnetic poles of Karelia into the reference frame of Superior using the continental configuration Superia by Euler pole of ELat = 75.5°, ELon = 251° and rotation angle = −114° (Fig. 10). In this configuration coeval paleomagnetic poles of these cratons are not overlapping, which indicates that a Superia fit is not permitted by paleomagnetic data. However, a closer fit between these continents is supported by 2.5 and 2.1 Ga paleomagnetic data (Figs. 9 and 10). According to paleomagnetic data Karelia occupied equatorial latitudes at 2.5 Ga moving on to the intermediate latitudes at 2.45 Ga and back to the equator at 2.33 Ga when it acquired the magnetization component *D′*. At 2.07 Ga it was located on latitudes of 20–25°. At the same time, Superior occupied latitudes of 20–25° at 2.5 Ga and at 2.45 Ga. It moved to higher latitudes at 2.2 Ga and reached the latitude of ca. 40° at 2.17 Ga before returning to a latitude of 32° at 2.07 Ga (Fig. 9). Differences in latitudinal drift would imply that these continents do not share a joint history at ca. 2.45 Ga and therefore are not supportive of the close reconstruction of Bleeker and Ernst (2006). The shallow inclination remanence obtained from the baked contact rock of the 2.33 Ga dyke (AD13) is seen only in this single dyke in Taivalkoski, but was previously seen in two dykes in Lake Pääjärvi area in Russian Karelia (Mertanen et al., 1999). This remanence was recently used by Mertanen and Pesonen (2012) and Pesonen et al. (2012) to represent 2.45 Ga magmatism (Paa D′, 2450 Ma in Fig. 9) in Karelia, a result consistent with the Karelia-Superior reconstruction shown by Bleeker and Ernst (2006). The reason for using this pole was its resemblance with the pole from dyke AD13, originally thought to be about 2.4 Ga old, but which has now turned out to be 2.33 Ga. Therefore, we now consider that the interpretation by Mertanen and Pesonen (2012) and Pesonen et al. (2012) is invalid as the pole from the AD13 dyke is clearly younger than 2.45 Ga. Here we suggest that the Burakovka, Paa D, and the new Taivalkoski D pole are most likely primary 2.45–2.4 Ga poles, verified by baked contact samples on dyke WD dated by Sm–Nd at 2407 ± 35 Ma. The robustness of Sm–Nd model age for dyke WD has been discussed above and the ages for Burakovka and Paajarvi intrusions are obtained using robust U–Pb method (Table 3). The unity of Superior and Karelia cratons at 2.5–2.1 Ga is anyhow supported by geological and geochemical similarities, but the connection at 2.45 Ga still remains to be tested.

5. Conclusions

The main conclusion from the new paleomagnetic data and age dating is that in the Karelia Province the E–W trending dyke swarm with an age of 2.33 Ga shows its own characteristic SE pointing low inclination direction (component *D′* from dyke AD13), and a more NW–SE trending 2.45 Ga dyke swarm has its own SE pointing intermediate inclination direction (component *D* from dyke WD). Provided that these paleomagnetic data are primary the Karelia and Superior Provinces were not attached at 2.4 Ga or at 2.33 Ga but were in close proximity at 2.5 Ga and 2.1 Ga, thus negating the proposed Superia-fit based upon dyke swarm trajectories.

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